# Notes for Chapter 2

## Determinants

### 2.1: The Determinant of a Matrix

#### Definition of Minor and Cofactor:

Let $A = (a_{ij})$ be an $n \times n$ matrix, and let $M_{ij}$ denote the $(n-1) \times (n-1)$ matrix obtained from $A$ by deleting the row and column containing $a_{ij}$. The determinant of $M_{ij}$ is called the **minor** of $a_{ij}$. We define the **cofactor** $A_{ij}$ of $a_{ij}$ by

$$A_{ij} = (-1)^{i+j} \det(M_{ij})$$

#### Determinant Calculation:

The **determinant** of an $n \times n$ matrix $A$, denoted $\det(A)$, is a scalar associated with the matrix $A$ that is defined inductively as

$$\det(A) = \begin{cases} a_{11} & \text{if } n = 1 \\ a_{11}A_{11} + a_{12}A_{12} + \cdots + a_{1n}A_{1n} & \text{if } n > 1 \end{cases}$$

where

$$A_{1j} = (-1)^{1+j} \det(M_{1j}) \qquad j = 1, \ldots, n$$

are the cofactors associated with the entries in the first row of $A$.

#### Theorem 2.1.1:

If $A$ is an $n \times n$ matrix with $n \geq 2$, then $\det(A)$ can be expressed as a cofactor expansion using any row or column of $A$; that is,

$$\begin{aligned} \det(A) &= a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in} \\ &= a_{1j}A_{1j} + a_{2j}A_{2j} + \cdots + a_{nj}A_{nj} \end{aligned}$$

for $i = 1, \ldots, n$ and $j = 1, \ldots, n$.

***Note:*** The above theorem illustrates that the determinant can be calculate by *either a row or a column*.

#### Theorem 2.1.2:

If $A$ is an $n \times n$ matrix, then $\det(A^T) = \det(A)$.

#### Theorem 2.1.3:

If $A$ is an $n \times n$ triangular matrix, then the determinant of $A$ equals the product of the diagonal elements of $A$.

#### Theorem 2.1.4:

Let $A$ be an $n \times n$ matrix.

**(i)** If $A$ has a row or column consisting entirely of zeros, then $\det(A) = 0$.

**(ii)** If $A$ has two identical rows or two identical columns, then $\det(A) = 0$.

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## *2.2: Properties of Determinants*

### Lemma 2.2.1:

Let $A$ be an $n \times n$ matrix. If $A_{jk}$ denotes the cofactor of $a_{jk}$ for $k = 1, \ldots, n$, then

$$a_{i1}A_{j1} + a_{i2}A_{j2} + \cdots + a_{in}A_{jn} = \begin{cases} \det(A) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

### Determinants of Elementary Matrices:

In summation, if $E$ is an elementary matrix, then

$$\det(EA) = \det(E)\det(A)$$

where

$$\det(E) = \begin{cases} -1 & \text{if } E \text{ is of type I} \\ \alpha \neq 0 & \text{if } E \text{ is of type II} \\ 1 & \text{if } E \text{ is of type III} \end{cases} \tag{2}$$

Similar results hold for column operations. Indeed, if $E$ is an elementary matrix, then $E^T$ is also an elementary matrix (see Exercise 8 at the end of this section) and

$$\begin{aligned} \det(AE) &= \det\left((AE)^T\right) = \det\left(E^T A^T\right) \\ &= \det\left(E^T\right)\det\left(A^T\right) = \det(E)\det(A) \end{aligned}$$

### Theorem 2.2.2:

An $n \times n$ matrix $A$ is singular if and only if

$$\det(A) = 0$$

### Theorem 2.2.3:

If $A$ and $B$ are $n \times n$ matrices, then

$$\det(AB) = \det(A)\det(B)$$

### Lemma 2.2.4:

If one row of the determinant of A is equal to the sum of two numbers, then the determinant of A equals to the sum of two determinants. It follows that

$$\begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ b_1+c_1 & \cdots & b_n+c_n \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix} = \begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ b_1 & \cdots & b_n \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix} + \begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ c_1 & \cdots & c_n \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix}$$

**Lemma 2.2.5:**

If two rows of the determinant of matrix A are proportioned, then det(A)=0.